Multi-Source ALT Classification in Osteosarcoma

A Computational Feasibility Study

Rinat Rizvanov · June 2026 · Independent computational analysis

Agenda

- Scientific motivation — OS, ALT, TCAB1
- Research aims and go/no-go framework
- Pipeline architecture and data sources
- Labeling framework — four-tier hierarchy
- Feasibility audit — Pass A and Pass B
- RNA availability crisis and three recovery routes
- Aim 2 discovery attempt — a null result (Notebook 11)
- Sex confounding diagnosis (Notebook 12, D1 & D2)
- Hypothesis-driven curated gene score — D3
- Robustness analysis (Notebook 13)
- Limitations and future directions

Scientific Background

Why osteosarcoma, ALT, and TCAB1?

Osteosarcoma: The Problem

- Most common primary bone malignancy in children and adolescents
- 5-year survival: ~65–70% localized / <20% metastatic or relapsed
- Standard of care = MAP chemo (methotrexate, doxorubicin, cisplatin) — unchanged since 1980s
- Extreme genomic instability: near-universal TP53 disruption, complex SVs
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- Defining molecular feature:
  - ~50% of osteosarcomas use Alternative Lengthening of Telomeres (ALT)
  - instead of telomerase — highest ALT prevalence of any tumor type studied
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- ALT makes OS an ideal system to study ALT as a therapeutic vulnerability

What Is ALT?

Biology

- Homologous-recombination-based telomere maintenance
- Replaces telomerase in ~50% of osteosarcomas
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- Molecular hallmarks:
  - ALT-associated PML bodies (APBs)
  - C-circle assay — extrachromosomal telomeric repeat DNA
  - Loss of ATRX/DAXX → aberrant H3.3 loading at telomeres
  - Heterogeneous, ultra-long telomeres

Therapeutic relevance

- ALT cells depend on:
  - HR machinery for telomere synthesis
  - FANCM-mediated break-induced replication at telomeres
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- These dependencies are NOT shared by telomerase+ cells
- → Potential synthetic lethal targets
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- Experimental assays for ALT:
  - C-circle assay, APB immunofluorescence, TRAP (telomerase activity)

TCAB1 / WRAP53 — The Original Question

- TCAB1 (Telomerase Cajal Body protein 1), encoded by WRAP53 at 17p13.1
- Localizes telomerase to Cajal bodies — required for efficient telomere extension
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- In an ALT-positive tumor:
  - TCAB1 function may be irrelevant or actively harmful
  - Evidence: TCAB1-disrupted tumors may depend more on ALT-mediated repair
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- Two research questions:
  - 1. Is TCAB1 genomic disruption (17p13.1 SV breakpoints) associated
  - with a distinct transcriptional program in osteosarcoma?
  - 2. Can a transcriptional signature distinguish ALT+ from ALT− lines?

Pipeline Architecture & Data

What I built and where the data came from

Repository Architecture

Structure

- src/ — all reusable logic
  - manifest_builder.py — cohort loaders
  - label_rules.py — deterministic labels
  - sv_annotator.py — 17p13.1 SV calls
  - audit.py — Pass A / B logic
  - aim2_signature.py — LOOCV training
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- notebooks/ 00–13 — thin orchestration layers
- tests/ — 26 unit tests, all passing
- data/manifests/ — master_manifest.csv (115×46)
- data/processed/ — reports, sha256 hashes

Principles

- Design principles:
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- Notebooks import from src/ —
  - one fix propagates everywhere
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- Every output is sha256-verified —
  - byte-identical re-runs confirmed
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- Analysis spec frozen in label_spec_v1.0.md
  - before any modeling code was written
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- Prevents circular method refinement

Five Cohorts, 115 Samples

Cohort	N	Label type	RNA-seq
DepMap 26Q1	35 cell lines	ALTitude predictions	✅ 37 with RNA
PCGP 2014	34 pediatric tumors	SV + TP53 (peer-reviewed)	✗ public release
de Nonneville 2022	22 bone tumors	C-circle / APB (peer-reviewed)	✗ microarray only
Lawlor 2026 GCCRI	17 PDX	SV / IHC (preprint)	✗ internal only
Taglialatela 2026	7 cell lines	ALT list (preprint)	✗

Key tension: Labels and expression data are in completely disjoint datasets.
The DepMap RNA matrix (1,719 cell lines × 19,177 genes) had no ALT labels.
The labeled cohorts had no RNA-seq available in their public releases.

Labeling Framework

Four-tier hierarchy, frozen before any modeling

ALT Status Hierarchy

Tier	Label	Source	Used in
1	alt_gold_positive	C-circle / APB / TRAP, peer-reviewed	Primary + Pragmatic
1	alt_negative_high_conf	Same experimental evidence	Primary + Pragmatic
2	alt_silver	ALTitude prediction only (no wet-lab confirmation)	Pragmatic only
3	alt_unknown	No usable evidence	Excluded from all modeling

- Three parallel definitions: Strict (gold only) → primary | Pragmatic (gold+silver) → sensitivity | Broad → no new samples
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- Critical: alt_call_direction (positive/negative/unknown) is the arm-assignment column
  - Silver labels include BOTH ALT+ (n=6) and ALT− (n=5) predictions
  - Using alt_status directly would assign all silver to the positive arm — wrong
  - alt_call_direction was added in Notebook 10 after recognizing this during the merge

Feasibility Audit

Two checks before any modeling code was written

Pass B (Confounding) + Pass A (Availability)

- Pass B: Confounding Census
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- Checks if confounders are imbalanced
- across TCAB1 and ALT arms
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- Critical finding (Aim 1):
  - PCGP 2014 = 100% of TCAB1-intact arm
  - Any DE result would be completely
  - confounded with cohort of origin
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- Documented: Halt override: True
- Audit runs to completion for records

- Pass A: RNA Availability Census
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- Checks how many labeled samples
- have file-backed RNA-seq on disk
-
- Initial result:
  - Aim 1 RNA-backed disrupted = 0
  - Aim 2 RNA-backed ALT-labeled = 0
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- Stage 0 diagnostics (nb06, nb07)
- were designed to catch this before
- any analysis code was written
- → not a failure, a designed check

RNA Crisis and Recovery

Three routes — one succeeded

Three Recovery Routes

Route	Description	Outcome
A	de Nonneville → ArrayExpress E-MTAB-11743	Affymetrix microarray (not RNA-seq) → external validation only
B ✅	ALTitude predictions → DepMap cell-line-name join	Succeeded: 37 RNA-backed OS lines matched → Aim 2 rescued
C	Lawlor PDX internal RNA-seq (PI collaboration)	Pending — critical dependency for Aim 1

- The column-binding bug (Notebook 09):
  - Initial run: NOT ESTIMABLE — suspicious given 37 RNA-backed lines
  - Root cause: code read Curated.Marker (ATRX mutation feature) not ALT_pred_class
  - Caught by ground-truth check: U2OS returned ambiguous_or_missing → wrong column
  - Fixed; corrected inventory: 31 RNA-backed lines / 12 ALT+ / 7 ALT− gold
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- After Notebook 10 direction-aware merge:
  - Primary (gold): N = 20 (12 ALT+ / 8 ALT−) Pragmatic (gold+silver): N = 31 (18 ALT+ / 13 ALT−)

Aim 2 Modeling

Notebook 11 → 12 → 13

Notebook 11: Whole-Transcriptome Discovery — Null

- Nested LOOCV, elastic-net logistic regression, 19,177 genes
- Every preprocessing step is fold-local (no leakage):
  - VarianceGeneFilter (top K=500) → StandardScaler → ElasticNet (inner 5-fold C selection)

Definition	N	AUROC [95% CI]	Permutation p
Primary (gold)	20	0.4688 [0.20, 0.75]	0.60 (null)
Pragmatic (gold+silver)	31	0.6880 [0.45, 0.90]	0.06 (borderline)

- Primary = worse than random chance
- Top features exposed the problem:
  - Ranks 1–4: XAGE1B, MAGEB2, XAGE1A, MAGEA10 (X-linked cancer-testis antigens)
  - Ranks 17–18: USP9Y, DDX3Y (Y-chromosome genes)
  - Zero ALT-gene-list genes in top 20 of either definition
- The model was learning sex, not ALT biology

Notebook 12: Sex Confounding Diagnosis

Design	Method	Primary AUROC	Perm p
D1 — sex covariate	Append Female/Male indicator after gene filter	0.5000	0.53
D2 — no Y/XCT genes	Remove 41 Y-chr + 4 X-CT antigen columns	0.3958	0.66
D3 — curated ALT genes	19-gene prespecified list, no variance filter	0.9479	< 0.001

- D1 and D2 confirm: the borderline pragmatic signal in nb11 depended entirely on sex-linked features
- After sex correction: no ALT signal survives the unguided whole-transcriptome approach
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- D3 is a different kind of test — a hypothesis test, not a discovery attempt
  - Feature set fixed before any model fitting using a pre-specified biological gene list
  - Bypasses the problem of finding a needle in 19,177-gene haystack at N=20

D3: Hypothesis-Driven Curated Gene Score

- 21-gene ALT pathway list: TERT, TERC, ATRX, DAXX, SMARCAL1, TP53, WRN, BLM, MRE11A,
  - RAD50, NBN, FANCD2, RAD51AP1, WRAP53, RAD52, BRCA1, BRCA2, FANCM, SLX4, GEN1, MUS81
- 19 of 21 found in DepMap matrix (TERC = ncRNA; MRE11A = symbol mismatch in DepMap)

Definition	N	AUROC [95% CI]	AUPRC	Sens@90%Spec	Perm p
Primary (gold)	20	0.9479 [0.81, 1.00]	0.9755	91.7%	< 0.001
Pragmatic	31	0.8291 [0.67, 0.96]	0.9166	77.8%	0.010

- Primary nonzero coefficients (full-data refit, C = 0.3162):
  - TERT −0.689 FANCM +0.683 BRCA2 +0.197 WRN −0.153 NBN +0.105
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- Important framing: this is a test of a pre-formed hypothesis, not a discovery
  - The result says these 19 genes carry information about ALT status in this cohort
  - It does not establish generalization to new cohorts — external validation is needed

Gene Interpretation: TERT and FANCM

- TERT (negative coefficient)
- Catalytic subunit of telomerase
- Telomerase and ALT are mutually exclusive
- → Lower TERT expression = ALT-positive
- Most stable predictor across all analyses:
  - Bootstrap freq = 1.00 (both definitions)
  - LOOCV fold freq = 1.00 (all three cohorts)
- WRAP53 (zero coefficient)
- Central to the original question, but
- WRAP53 mRNA adds no information
- beyond TERT + FANCM at N = 20

- FANCM (positive coefficient)
- Fanconi anemia helicase required for
- break-induced replication at ALT telomeres
- Bhowmick et al. (2016): FANCM depletion
- selectively inhibits ALT activity
- → Proposed synthetic lethal target in ALT+
- Higher FANCM expression = ALT-positive
- Bootstrap freq = 1.00 (both definitions)
- Second most stable predictor overall

Notebook 13: Robustness Analysis

Cohort	N	AUROC	Perm p (1,000 shuffles)
Primary	20	0.9479	< 0.001 (0/1,000)
Pragmatic	31	0.8291	0.010 (10/1,000)
Pragmatic, known-sex only	27	0.8765	0.003 (3/1,000)

- Extended permutation test (1,000 shuffles vs 100 in nb12):
  - Primary highly robust (0/1,000); pragmatic nominally significant — interpret cautiously
  - Implementation: precomputed fold-local StandardScaler once (label-independent) → ~10× faster
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- Known-sex-only pragmatic (drop 4 unknown-sex samples):
  - AUROC improves 0.8291 → 0.8765; CI tightens; p = 0.003
  - TP53 coefficient shrinks substantially (−0.337 → +0.101) when unknown-sex dropped
  - Suggests TP53's pragmatic contribution was entangled with label uncertainty

Bootstrap Coefficient Stability (200 Resamples)

Gene	Primary freq	Primary class	Pragmatic freq	Pragmatic class
TERT	1.00	stable	1.00	stable
FANCM	1.00	stable	1.00	stable
BRCA2	0.81	stable	0.655	borderline
TP53	0.445	unstable	0.935	stable
WRN	0.57	borderline	0.705	borderline
NBN	0.625	borderline	0.785	borderline
ATRX	0.235	unstable	0.555	borderline
12 others	< 0.30	unstable	—	—

TERT and FANCM: the only genes stable in both definitions. LOOCV fold frequency confirms both appear in every held-out fold (freq = 1.00). The result is not driven by any single sample.

Limitations

- Sample size
  - Primary N = 20 (12+ / 8−)
  - AUROC CI = [0.81, 1.00] — wide
  - Expected AUROC in new samples
  - could be substantially lower
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- Label quality
  - Gold labels: different groups,
  - different protocols, different studies
  - Silver (ALTitude): computationally derived,
  - never independently validated
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- Single platform
  - DepMap 26Q1 log2(TPM+1) only
  - Cell lines ≠ primary tumor profiles

- This is confirmatory, not discovery
  - D3 tests prior biological knowledge
  - High AUROC validates the prior —
  - it does not discover new biology
  - Framing it as novel signature discovery
  - would overstate the finding
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- Permutation test conservatism
  - C fixed at median observed value
  - (fully nested null is intractable ~9h)
  - Conservative: true p may be smaller
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- No probability calibration
  - LOOCV scores are not calibrated probabilities
  - This is a signature score, not a classifier

Future Directions

Priority	What	Dependency
Near-term	External validation on de Nonneville microarray (22 tumors, C-circle labels) Apply 19-gene curated score after within-platform normalization	None — can pursue independently
PI collaboration	Aim 1: Lawlor PDX RNA-seq 8 TCAB1-disrupted vs 32 intact → differential expression	Internal RNA-seq from Lawlor 2026 GCCRI group
DepMap only	FANCM dependency in DepMap CRISPR (Chronos scores) Does FANCM dependency correlate with ALT status?	None — data already on disk
Alternative Aim 1	WRAP53 expression-quantile contrast within DepMap Stratify 35 OS lines by WRAP53 expression level	None — but answers a different question than Aim 1

Summary

What was done	Result
5-cohort master manifest (115 samples, 46 columns)	Built and audited
Four-tier ALT label hierarchy, frozen spec	Documented in label_spec_v1.0.md
Stage 0 RNA availability audit	Both aims initially inestimable
Three recovery routes evaluated	Route B rescued Aim 2
Column-binding bug caught and fixed	U2OS sanity check
Aim 2 whole-transcriptome LOOCV (nb11)	NULL — sex confound identified
Sex confounding diagnosis (nb12 D1, D2)	Confirmed: no ALT signal after correction
Curated gene score, primary N=20 (nb12 D3)	AUROC = 0.948, $p < 0.001$
1,000-shuffle permutation test (nb13)	Primary 0/1,000; pragmatic 10/1,000
Bootstrap + LOOCV fold stability (nb13)	TERT and FANCM stable in all analyses
Aim 1 (TCAB1 differential expression)	NOT ESTIMABLE — pending Route C

Key Takeaway

The ALT transcriptional signal in osteosarcoma is real but concentrated in a small number of functionally motivated genes.
It cannot be found by unguided gene selection at N = 20,
but it is recoverable when you know where to look.

TERT and FANCM are robust across every analysis.

WRAP53 mRNA does not contribute to the classification signal.

Rinat Rizvanov · github.com/rinatrizvanov